Xedni: dependency-bound audit

2026-11-01 | TASK-20261101-fbd859 | H-XEDN-292264 (proposed)

Component progress

The square-root prime-divisor inference fails. A conditional integer majorant is derived separately; the complete repaired local-test theorem remains open. Historical diagnostic: 20 prime-divisor pairs, 10 square-root violations. Imported conversation evidence, not a harness RUN or independent replay.

Supplied rational data

E/Q, full basis, P and Gamma

Coefficient bound

Nonzero n: prime ell divides n; |n| <= C

Invalid step

ell <= sqrt(C): fails at n = ell = C = 3

Conditional repair

ell <= B and k <= K imply degree <= M_star

Open proof bridge

Admissible k, fields, zero/unit cases, full cutoff

No reverse lift construction is supplied by this reduction-based test.

Source: Sadek, arXiv:1605.02961v1, 2016-05-10, Theorem 4.4 proof.
Published: C. R. Math. Rep. Acad. Sci. Canada 38(2) (2016), 75-84.
https://arxiv.org/pdf/1605.02961 | https://mathreports.ca/download/1173/

Repair hypothesis and next obligations

Conditional arithmetic lemma

For integers 2 <= ell <= B, 1 <= k <= K and r >= 1:

(ell^2-1)(ell^2-ell) ell^(2(k+1)r+4k)

<= (B^2-1)(B^2-B) B^(2(K+1)r+4K) = M_star.

Proof: positive monotone factors, base >= 2, increasing exponent.

This does not prove that an admissible uniform integer K exists.

H-XEDN-292264: proposed full repair

Reconstruct a valid cutoff with certified B, admissible K and M_star.

Retain all Galois, GRH, full-basis and exceptional-j hypotheses.

Independently audit Kummer fields, integer rounding and zero/unit cases.

The reported gap is not a disproof of the local-to-global principle.

Next bounded check - not launched

B=2..64, r=1..4, K=1..8; all prime ell<=B and integer k=1..K.

Exact integer comparisons; equality and invalid-domain controls.

Retain every tuple, values, exceptions and full resource measurements.

Freeze protocol and pass checkpoint/run admission before a science sweep.

Acceptance and cost accounting

Local success is a filter result, not an exact rational relation certificate.

Charge lift search, failures, basis/height work, local tests, coefficient discovery, exact verification, recovery, memory and reusable setup.

Unique scalar recovery from aP+bQ=O needs gcd(b,N)=1.

Historical V2 comparison (not rerun)

68 doubling checks: one residue per prime. 1,944 rational fits / 24 pairs; 20 pairs had finite-field coefficients in [-8,8]; zero certified rational dependencies in that bound. This audit does not change those results.

Tracking and evidence

README.md: sources, scope, costs and obligation table.

prior-command.txt / prior-output.txt: actual earlier invocation/output.

test_bounds.py: software controls; VALIDATION.md: observed checks.

problems/xedni-sadek-explicit-cutoff.md: unresolved proof requirements.